

Analytics Methodology & Data Schema

NCIIPC — Problem Statement 26157

SAT-SA helps NCIIPC supervisors identify which Critical Sector Entities, processes and cases warrant closer review — by turning periodic SOC operational evidence into explainable, evidence-backed supervisory signals. It supports human examiners; it does not replace their judgement.

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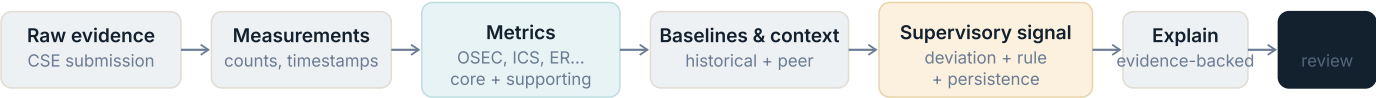
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01 The analytics approach

A metric is a measurement. It is not, by itself, a supervisory finding. SAT-SA moves from raw evidence to a human decision through a deliberately staged chain — each stage adds context before anything is called a signal.



Why core and supporting metrics are kept separate

Every metric in this specification belongs to one of two classes. The split exists so that a supervisory finding is never generated by a single thin or context-only measurement acting

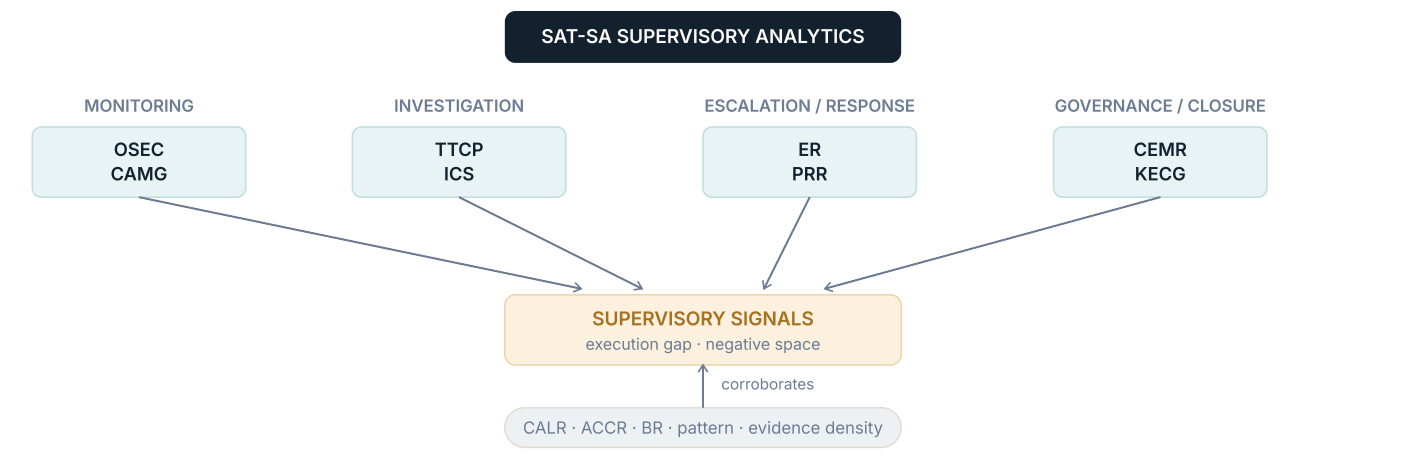
alone — which matters directly for explainability and for validating SAT-SA against expert manual review.

CORE METRICS

May, on their own or in combination, produce a candidate execution-gap or negative-space signal.

SUPPORTING / CONTEXTUAL METRICS

Never trigger a finding alone. They corroborate, weight or explain a core signal — e.g. distinguishing capacity strain from disregard.



02 Core supervisory metrics

Eight metrics. Each is defined with its formula, the data it requires, how to read it, and a worked (illustrative) example. Every core metric is tagged by the kind of signal it can contribute to.

OSEC Observed Security Evidence Coverage negative space

Proportion of assets expected to produce security evidence that have at least one qualifying evidence record in the assessment period.

OSEC = (assets with expected evidence observed ÷ assets expected to produce evidence) × 100

EXAMPLE

40 assets expected to log evidence; 34 have a qualifying record → OSEC = 34/40 × 100 = 85%

READING IT

Low OSEC flags a possible visibility gap — it does not by itself prove monitoring is absent.

CAMG Critical-Asset Monitoring Gap negative space

Proportion of *critical* assets expected to be monitored that show no qualifying evidence in the period.

$$\text{CAMG} = (\text{critical assets without qualifying evidence} \div \text{critical assets expected to be monitored}) \times 100$$

EXAMPLE

12 critical assets; 2 show no evidence → $\text{CAMG} = 2/12 \times 100 \approx 16.7\%$

READING IT

High CAMG marks a critical-asset blind spot worth review — a negative-space candidate, not a confirmed failure.

TTCP Time-to-Close Percentile execution gap

Where a case's time-to-close falls within a comparable population (same CSE, alert type, severity and period).

$$\text{TTC} = \text{closed_at} - \text{opened_at} \quad \text{TTCP} = \text{PercentileRank}(\text{TTC} \mid \text{comparable group})$$

EXAMPLE

A critical-alert case closes in 2 hours, while comparable cases typically take much longer → case lands near the 8th percentile (unusually fast).

READING IT

Unusually low TTCP should always be read together with ICS — speed alone is not evidence of quality.

ICS Investigation Completeness Score execution gap

How many of the applicable investigation requirements are actually evidenced in the record (alert reviewed, evidence examined, findings documented, escalation decision

documented, etc.).

$$\text{ICS} = [\sum(w_j \times I_j) \div \sum w_j] \times 100, I_j \in \{0,1\}, w_j = \text{weight of requirement } j$$

EXAMPLE

6 of 8 equally-weighted requirements satisfied $\rightarrow \text{ICS} = 6/8 \times 100 = 75\%$

READING IT

Low ICS means fewer expected investigation elements are evidenced in what was submitted.

ER Escalation Compliance Rate execution gap

Of the cases that meet defined escalation criteria, how many were actually escalated.

$$\text{ER} = (\text{qualifying cases actually escalated} \div \text{cases meeting escalation criteria}) \times 100$$

EXAMPLE

10 cases qualify for escalation; 6 were escalated $\rightarrow \text{ER} = 60\%$

READING IT

The denominator is qualifying cases only — not all cases handled in the period.

PRR Post-Remediation Recurrence Rate execution gap

Of the conditions recorded as remediated, how many recur — same asset, same/equivalent condition — within a configured window.

$$\text{PRR} = (\text{conditions recurring after recorded remediation} \div \text{conditions with recorded remediation}) \times 100$$

EXAMPLE

25 conditions remediated; 4 recur within the window $\rightarrow \text{PRR} = 16\%$

READING IT

High PRR identifies a recurring condition — it does not automatically prove the remediation failed.

CEMR Closure-Evidence Mismatch Rate execution gap

Of cases closed as resolved or remediated, how many lack the supporting evidence the submission would be expected to contain.

$$\text{CEMR} = (\text{resolved/remediated cases lacking supporting evidence} \div \text{resolved/remediated cases}) \times 100$$

EXAMPLE

50 cases closed as remediated; 9 lack supporting evidence → CEMR = 18%

READING IT

Required wording: *"No supporting evidence was found in the submitted dataset."*

Never phrase this as proof remediation didn't occur.

KECG KPI-Evidence Consistency Gap execution gap · governance

The discrepancy between what a CSE reports for a KPI and the value SAT-SA independently derives from the same evidence — only once both are aligned on definition, population, period and scope.

$$D_k = |\text{ReportedKPI}_k - \text{ObservedKPI}_k| \quad \text{KECG} = \sum (w_k \times D_k) \div \sum w_k$$

EXAMPLE

Reported closure-rate KPI = 95%; SAT-SA-observed = 81% → $D_k = 14$, contributing to a composite KECG once combined with other KPIs.

READING IT

High KECG means reported performance materially differs from what the submitted evidence shows — not proof of manipulation.

03 Supporting / contextual metrics

These never generate a finding on their own. They corroborate a core signal, adjust its priority, or supply the context needed to interpret it correctly.

METRIC	FORMULA	ROLE
CALR	<code>cases assigned ÷ active analysts</code>	Workload context for investigation/closure patterns
ACCR	<code>alerts linked to cases ÷ alerts × 100</code>	Triage/workflow context — not inherently good or bad
BR	<code>open cases at period end ÷ cases opened</code>	Backlog / operational-capacity context
Investigation Pattern Concentration	<code>actions in dominant pattern ÷ total actions × 100</code>	Flags repetitive investigation workflows; read by case type
Evidence Sufficiency / Density	<code>evidence items ÷ applicable requirements</code>	Context for TTCP, ICS, CEMR — quantity ≠ quality

04 Composite prioritisation — CEGS

The Composite Execution-Gap Score exists to rank the worklist, not to issue a verdict on SOC quality.

$CEGS_e = \sum(w_k \times S_{e,k})$ — $S_{e,k}$ = normalised strength of signal k for entity e;
 w_k = configurable weight

Inputs may include monitoring, investigation, escalation, recurrence, closure-evidence and KPI-evidence signals. Weights live in SAT-SA's configuration — never in the CSE's own submission. A higher CEGS means higher review priority; it is *not* a definitive compliance judgement.

05 Signal generation

`Signal = f(Metric, HistoricalBaseline, PeerBaseline, Threshold, Context)`

In sequence: `metric` → `sample sufficiency` → `historical/peer baseline` → `deviation` → `configurable rule` → `persistence/trend` → `contextual check` → `signal` → `evidence` → `human`

review.

SIGNAL	CONDITION
Potential superficial investigation	unusually low TTCP + low ICS, with sufficient sample
Potential monitoring blind spot	low OSEC + high CAMG, with historical/peer deviation
KPI-evidence discrepancy	elevated KECG, with aligned definitions and sufficient evidence
Post-remediation recurrence	elevated PRR, sufficient recurrence sample, persistent trend

Data sufficiency is not the same as poor performance. "ICS is low despite sufficient investigation records" is a performance signal. "ICS cannot be reliably assessed because investigation records are missing" is a data-sufficiency limitation. SAT-SA must always distinguish the two before a finding is raised.

06 Data schema

The CSE submits underlying operational evidence only. SAT-SA independently derives every metric, baseline, signal, explanation and priority score — nothing analytical is taken from the CSE at face value.

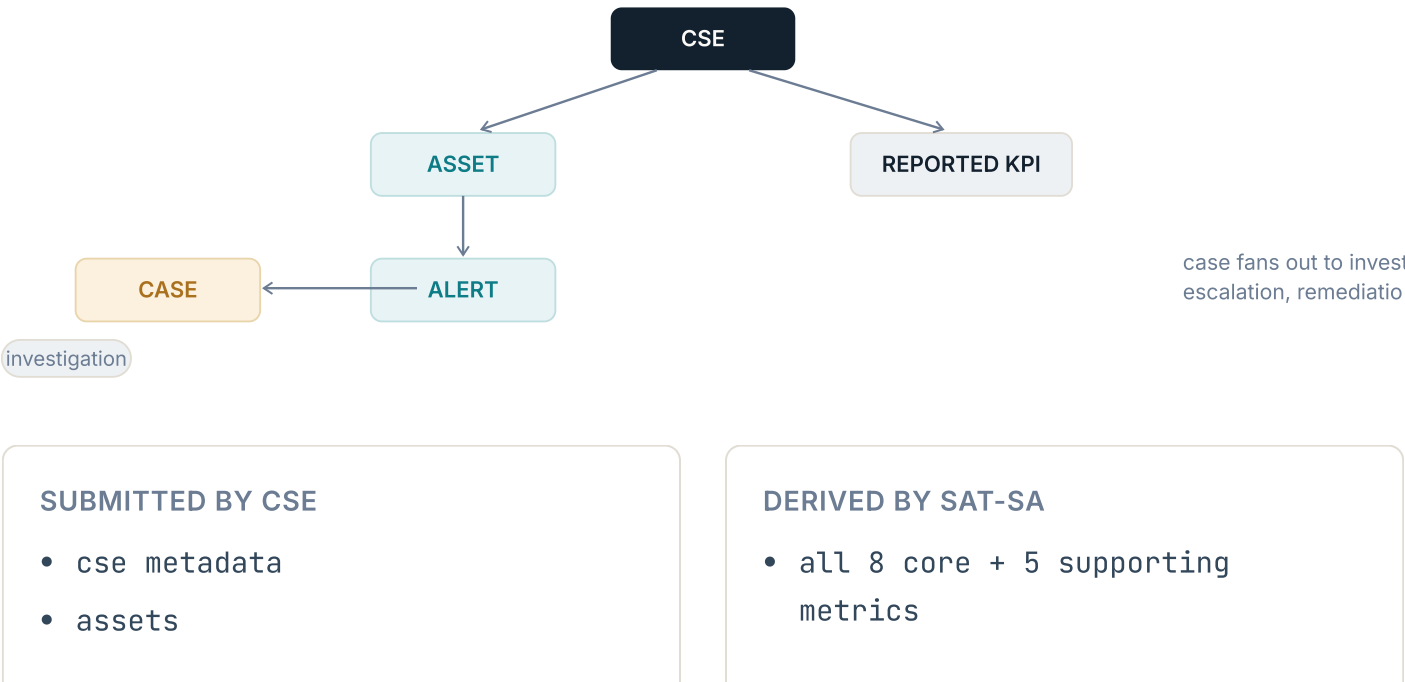
Submission envelope

```
submission
├─ submission_metadata
├─ cse
├─ assets[]
├─ alerts[]
├─ cases[]
├─ investigation_actions[]
├─ escalations[]
├─ remediations[]
├─ evidence[]
└─ reported_kpis[]
```

Entities at a glance

ENTITY	KEY FIELDS	SUPPORTS
asset	asset_id, criticality, monitoring_expected	OSEC, CAMG
alert	alert_id, asset_id, severity, created_at, status	TTCP, ACCR
case	case_id, alert_id, opened_at, closed_at, disposition	TTCP, ER, BR, CEMR
investigation_action	case_id, action_type, finding_recorded, disposition_recorded	ICS, pattern concentration
escalation	case_id, escalation_required, escalated_flag, escalated_at	ER
remediation	case_id, asset_id, remediation_recorded, remediation_status	PRR, CEMR
evidence	evidence_id, case_id, evidence_type, event_timestamp	OSEC, CAMG, ICS, CEMR, explainability
reported_kpi	kpi_id, reported_value, definition_version, population_definition	KECG

Entity relationships



- alerts
- cases
- investigation_actions
- escalations
- remediations
- evidence
- reported_kpis

- historical & peer baselines
- deviations & trends
- supervisory signals
- explanations
- CEGS
- prioritised worklist

End-to-end flow

CSE operational evidence → schema validation → entity resolution
→ raw + normalized storage → measurements → SAT-SA metrics
→ historical / peer baselines → supervisory signals
→ evidence-backed explanations → prioritized worklist → human examiner

07 Extending the model

This methodology and schema are intentionally non-exhaustive. The eight core metrics, five supporting metrics and nine schema entities documented here cover the data environment described in the problem statement — they are a starting set, not a ceiling.

New metrics are added as configuration — a formula, its required fields, a signal-type tag and an explanation template — through SAT-SA's metric registry, without changing the Analysis Engine, the Signaling Module, or any other core processing logic. New schema entities or fields extend the same way: additive, versioned, and validated before promotion to active use. The system is designed to grow with NCIIPC's supervisory understanding, not to be re-engineered every time it does.